

THE GREAT FRACTION FEAST: MAKING PIECES MATCH!

Adding and Subtracting Fractions with Different Bottom Numbers • Grade 4 Guide

What happens when slice sizes are different?

Let's learn how to turn different fractions into matching twins so we can add and subtract them easily!

Math Adventures: Adding & Subtracting Fractions!

The Great Barfi Sweets Problem!

Imagine a plate of sweet, delicious coconut Barfi sitting right in front of you!

Let's meet two friends, **Dev** and **Kiran**. They have some yummy Barfi pieces, and they want to put them together to find out how much they have in total.

- **Dev says:** "I have $\frac{1}{2}$ (one-half) of a tray of Barfi!"
- **Kiran says:** "I have $\frac{1}{3}$ (one-third) of a tray of Barfi!"

They try to add them up quickly. Dev says, "Let's just add the top numbers: $1 + 1 = 2$. And let's add the bottom numbers: $2 + 3 = 5$. So we have $\frac{2}{5}$!"

But wait! Kiran looks at the plate and pauses. $\frac{2}{5}$ is smaller than a half! How can putting their sweets together give them a smaller amount than what Dev started with? Something is very wrong here!

Let's Think: Why doesn't regular addition work here? Let's solve this mystery on the next page!

The Non-Matching Slice Rule

The mystery is simple: **The pieces are different sizes!**

Think about it. Dev's tray is cut into 2 big chunks. Kiran's tray is cut into 3 smaller chunks. Trying to add them together directly is like trying to add 1 giant elephant and 1 tiny mouse—you can't just say you have 2 elephant-mice!

The Golden Math Rule: In fractions, the Bottom Number (Denominator) tells us the exact size of each slice. You can **NEVER** add or subtract fractions unless their bottom numbers match perfectly!

When the bottom numbers are different, it means the slice sizes are completely different. To add or subtract them, we must find a way to make the slices look exactly the same.

Understanding Your Fraction Parts

Let's do a quick refresher on what a fraction means before we do our magic matching trick:

The Top Number (Numerator)

This tells us how many slices you have in your hand. If you have $\frac{3}{6}$, it means you are holding 3 slices.

The Bottom Number (Denominator)

This tells us the total number of equal slices the whole item was chopped into. It defines the size of the slice!

Slicing Fact!

The bigger the bottom number, the smaller the slices become! A cake cut into 100 pieces has tiny crumb slices, while a cake cut into 2 pieces has giant halves.

Meet the Twin Fractions!

How do we make different slices match? We use a magic trick called **Twin Fractions (Equivalent Fractions)**!

Let's imagine Dev and Kiran change their minds and take a round Indian Paratha. Dev has $\frac{1}{2}$ and Kiran has $\frac{1}{3}$.

To make them match, let's slice the whole paratha into 6 equal pieces total!

- Dev's big $\frac{1}{2}$ piece automatically turns into 3 smaller slices. So, $\frac{1}{2}$ is exactly the same size as $\frac{3}{6}$!
- Kiran's $\frac{1}{3}$ piece automatically turns into 2 smaller slices. So, $\frac{1}{3}$ is exactly the same size as $\frac{2}{6}$!

Look at that! $\frac{3}{6}$ and $\frac{2}{6}$ have the exact same bottom number! Now our slices match perfectly!

Adding Twin Fractions: Now It's Easy!

Now that Dev's and Kiran's fractions have been turned into matching twins, adding them is as simple as adding regular toys!

Let's look at our new matching math expression: $\frac{3}{6} + \frac{2}{6}$

Crucial Step Rule: When adding matching fractions, add **ONLY** the top numbers! The bottom number stays exactly the same because the size of the slices hasn't changed.

Math Solution: $3 + 2 = 5$, so the answer is $\frac{5}{6}$ of a tray!

Jalebi Treat: Subtraction Time!

We can use the exact same matching twin trick to subtract fractions! Let's look at a sweet example with Jalebi sweets.

Anjali has $\frac{3}{4}$ kg of juicy orange Jalebis. She wants to give her little brother $\frac{1}{2}$ kg of Jalebis. How much will she have left for herself?

Our math problem looks like this: $\frac{3}{4} - \frac{1}{2}$

Oh no! The bottom numbers (4 and 2) do not match. Let's make a twin fraction for $\frac{1}{2}$ so it matches the 4!

Multiply the top and bottom of $\frac{1}{2}$ by 2 to make it a twin fraction. It becomes $\frac{2}{4}$!

Now, let's subtract only the top numbers: $\frac{3}{4} - \frac{2}{4} = \frac{1}{4}$ kg of Jalebis left!

Guided Practice 1: Mixing Juices

Let's practice together with a refreshing fruit juice recipe!

Rahul is mixing juices in a large glass. He pours in $\frac{1}{3}$ of a glass of Mosambi (sweet lime) juice. Then, he adds $\frac{1}{6}$ of a glass of bright red Pomegranate juice.

What fraction of the glass is full of juice now?

- **Step 1: Write the problem:** $\frac{1}{3} + \frac{1}{6}$
- **Step 2: Match the bottoms!** Let's turn $\frac{1}{3}$ into sixths by multiplying top and bottom by 2. It becomes $\frac{2}{6}$!
- **Step 3: Add the matching twins!** $\frac{2}{6} + \frac{1}{6} = \frac{3}{6}$

Secret Bonus: $\frac{3}{6}$ is exactly half a glass! So, $\frac{3}{6} = \frac{1}{2}$ glass full! Amazing!

Guided Practice 2: Cricket Grip Tape

Let's do one more practice problem together, this time with a fun sports activity!

Aman has a roll of bright neon grip tape that is $\frac{5}{8}$ of a meter long. He cuts off a piece that is $\frac{1}{4}$ of a meter long to wrap around his favorite cricket bat handle.

How much tape is left on Aman's roll now?

- **Step 1: Write the problem:** $\frac{5}{8} - \frac{1}{4}$
- **Step 2: Match the bottoms!** Let's convert $\frac{1}{4}$ into eighths by multiplying the top and bottom by 2. It becomes $\frac{2}{8}$!
- **Step 3: Subtract the top numbers!** $\frac{5}{8} - \frac{2}{8} = \frac{3}{8}$

Aman has exactly $\frac{3}{8}$ of a meter of tape left on his roll!

You Are a Fraction Champion!

Fantastic work today! Let's review our three golden rules to remember forever:

Rule 1: Never add or subtract fractions if the bottom numbers are different!

Rule 2: Multiply top and bottom to create matching Twin Fractions.

Rule 3: Add or subtract ONLY the top numbers. Leave the bottom number alone!

Final Challenge Quiz! Can you solve $\frac{1}{2} + \frac{1}{4}$ using your new math powers? (Hint: Turn $\frac{1}{2}$ into quarters first, then add! Share your answer with your teacher!)

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The Great Fraction Feast: Practice Worksheet

Use your fraction magic rules to answer the questions below. Remember to make the bottom numbers match perfectly before adding or subtracting!

Section A: Core Concepts

Question 1

In a fraction, what special name do we give to the top number, and what does it tell us?

Question 2

What special name do we give to the bottom number of a fraction, and why can you **never** add or subtract fractions unless this number matches?

Question 3

True or False: "The larger the denominator (bottom number) is, the larger each individual slice of the item becomes." Explain why.

Question 4

What are **Twin Fractions** (Equivalent Fractions), and how do you turn a fraction into its twin?

Section B: Quick Calculations

Turn one of the fractions into its matching twin first, then add or subtract only the top numbers!

Question 5

Solve the addition problem: $\frac{1}{2} + \frac{3}{8} = \underline{\hspace{2cm}}$

Show your work:

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Question 6

Solve the addition problem: $\frac{1}{3} + \frac{2}{6} = \underline{\hspace{2cm}}$

Show your work:

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Question 7

Solve the subtraction problem: $\frac{7}{10} - \frac{2}{5} = \underline{\hspace{2cm}}$

Show your work:

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Question 8

Solve the subtraction problem: $\frac{3}{4} - \frac{1}{8} = \underline{\hspace{2cm}}$

Show your work:

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Section C: Tasty Word Problems

Question 9

Meera has a small tray of Halwa. She eats $\frac{1}{4}$ of the tray in the afternoon and another $\frac{3}{8}$ of the tray after dinner. What total fraction of the Halwa tray has she eaten?

Question 10

Arjun bought $\frac{5}{6}$ kg of juicy laddoos. He gifted $\frac{1}{3}$ kg of them to his best friend. What fraction of a kilogram of laddoos does Arjun have left for himself?

Question 11

A recipe for a special summer punch calls for $\frac{1}{2}$ cup of mango juice and $\frac{1}{10}$ cup of lemon juice. When mixed together, what total fraction of a cup will the juice measure?

Question 12

Imagine Dev and Kiran want to share a full pizza. Dev says they can calculate how much is left by doing 1 full pizza minus $\frac{1}{3}$ of a pizza. (Hint: Think of a whole number 1 as a twin fraction like $\frac{3}{3}$).

How much pizza is left?
